



Quad-core AMD Opteron processor

lowing that, the team was confident of the success of the SCC. However, this chip seems more complex, with a much larger die size, system-level complexities and the challenges of 45nm physical design to boot. The success of this chip will make processors more scalable than you ever thought possible, and also accelerate multiple-core software research and advanced development.

According to the company, future laptops with processing capability of this magnitude could have 'vision' similar to humans who can see objects and motion as it happens and with high accuracy. Imagine, for example, a laptop with a 3-D camera and display giving you virtual dance lessons or showing you a 'mirror' of yourself wearing the clothes you want to buy online. Twirl and turn to watch how the fabric drapes and how its colour complements your skin tone. This kind of interaction could eliminate the need of keyboards, remote controls or

joysticks for gaming. Some researchers believe computers may even be able to read brain waves, so simply thinking about a command, such as dictating words, would execute it without speaking.

We can expect more breakthroughs from Intel's stable considering the amount of research being pumped into this space. "On the hardware side, we explore scalable multi-core architecture that integrate streamlined processor cores and accelerators using a fast, energy-efficient, modular core-to-core infrastructure; memory sharing and stacking to provide a high-bandwidth, flexible, cache and memory hierarchy that supports many simultaneous threads fairly and efficiently; and high-bandwidth input/output (I/O) and communications that balance the compute demands within the I/O and network demands within the platform power and cost budgets," lists Erranguntla.

The soft side of success

As far as multi-core computing goes, success depends more than half on the ability of the program to efficiently harness the immense parallel computing power of the processor. "To take advantage of the increasing number of cores, efficient load balancing of software is required. In addition, we need to identify and come up with programs and applications for these systems, for performance improvement. It is also important for OEMs to strike a balance between performance and power consumption. We at AMD

understand this and are working with our partners to address these issues," says Vamsi Krishna.

Intel too has taken several initiatives to educate developers on the specific challenges and techniques of parallel computing, in order to enable them to make good use of multi-core systems. Plus, Intel and Microsoft have partnered with academia to create two Universal Parallel Computing Research Centres (UPCRC) aimed at accelerating developments in mainstream parallel computing for consumers and businesses in desktop and mobile computing.

The multi-core related software research vision at Intel includes "model-based applications that use tera-scale capabilities to comprehend data, make smarter decisions, and make visual experiences look, act and feel real; parallel programming tools that empower the ordinary programmer to develop applications that use parallelism with scalability and performance, safety and reliability; and thread-aware execution environments that provide real-time performance and power management across cores and scale with increasing thread and core counts," informs Erranguntla. You can find details at <http://techresearch.intel.com/articles/TeraScale/1421.htm>

More than desktops and supercomputers

Multi-core processors are becoming an indispensable aspect of virtualisation too, especially in data centres.

Vamsi Krishna explains, "The near future of multi-core processors will see the technology enabling data centres to accomplish tasks more quickly and with greater energy efficiency. Areas such as virtualisation are also primed for a boost by multi-core processors. As AMD moves to 8- and 12-core platforms, far more virtual machines can be packed onto each physical server. By focusing heavily on power efficiency and virtualisation capabilities, businesses can now add performance and efficiency to their business without a significant cost."

Multi-core microcontrollers for cars, phones and more

Multi-core technology is no longer confined to high-performance computing. It has entered the devices that we use every day, in the form of multi-core microcontrollers that are being extensively used in embedded systems.

Several companies including Renesas Technologies, LSI Corporation, STMicroelectronics and Freescale Semiconductor have introduced multi-core microcontrollers in the recent past. While the multi-core chips initially introduced for embedded systems were aimed at image processing and other multimedia products with heavy processing loads, recent products are targeted at a broad range of general-purpose devices. These facilitate the true real-time response times required by real-time control systems, safety critical applications, etc, thanks to their ability to process instructions in parallel.